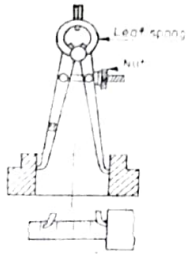
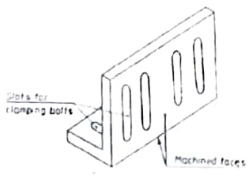
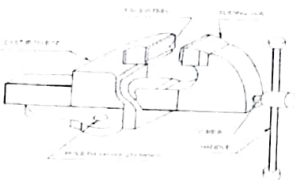
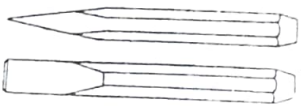
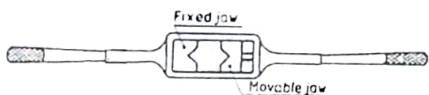
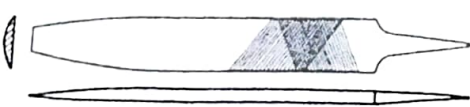
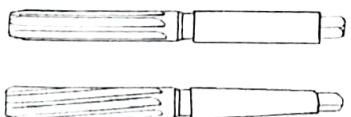
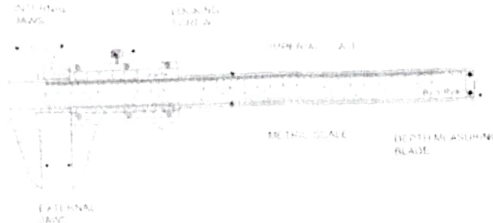
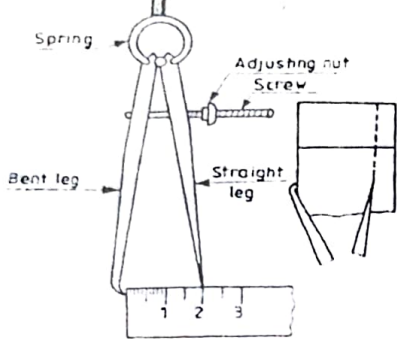
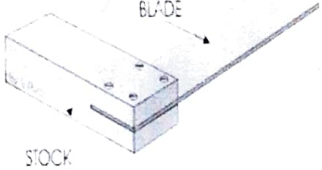
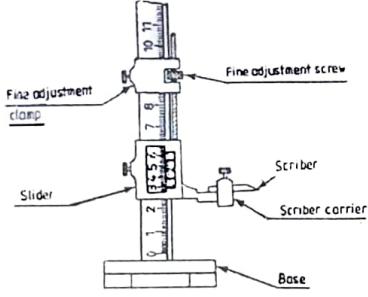
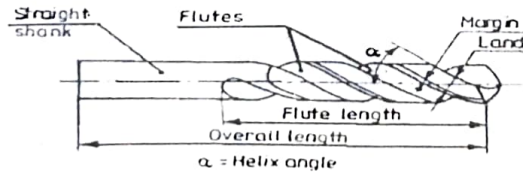


Aim 1: Identify the following fitting tools

| Sl. No | Tools                                                                               | Name of the tool | Type of Tool<br>(Holding tool/ Cutting tool/ Measuring Tool/<br>Finishing tools) |
|--------|-------------------------------------------------------------------------------------|------------------|----------------------------------------------------------------------------------|
| 1      |    |                  |                                                                                  |
| 2      |    |                  |                                                                                  |
| 3      |   |                  |                                                                                  |
| 4      |  |                  |                                                                                  |
| 5      |  |                  |                                                                                  |
| 6      |  |                  |                                                                                  |
| 7      |  |                  |                                                                                  |

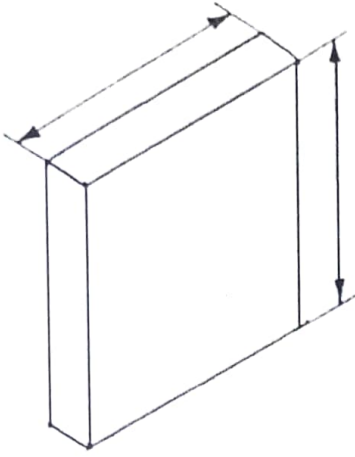
Name:

Enrollment No:

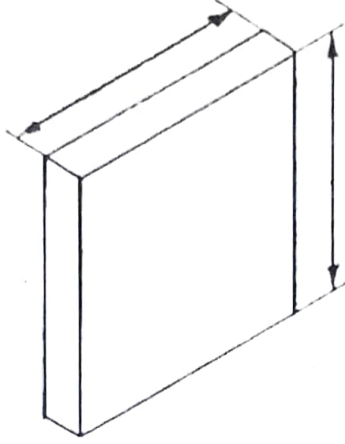
|    |                                                                                     |  |  |
|----|-------------------------------------------------------------------------------------|--|--|
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| 9  |    |  |  |
| 10 |   |  |  |
| 11 |  |  |  |
| 12 |  |  |  |

Name:  
Enrollment No:

**Aim 2: To file a given Mild Steel (M.S) piece into a square shape of given dimension**



a. Raw material



b. Finished job

**Figure: Square filing**

**Tools Required:**

- Bench Vice
- Set of Files (Rough and Smooth)
- Steel Rule
- Try-Square
- Vernier Caliper
- Vernier Height Gauge
- Ball-Peen Hammer
- Scriber
- Dot Punch
- Angle Plate
- Surface Plate

**Procedure:**

**1. Cutting the Mild Steel Strip:**

- The required size of the square mild steel strip is cut using a hacksaw from a longer steel strip.

**2. Checking Dimensions:**

- The dimensions of the cut piece are verified using a steel rule to ensure they are close to the desired size.

### **3. Filing Two Adjacent Sides:**

- The job is securely fixed in a bench vice.
- The two adjacent sides are filed using a rough flat file first, followed by a smooth flat file, ensuring that the two sides are at a right angle to each other.
- The right angle between the two adjacent sides is checked using a try-square.

### **4. Marking the Workpiece:**

- Apply chalk to the surface of the workpiece for better visibility.
- Mark the dimensions by scribing two lines, using the Vernier height gauge, with reference to the two datum sides. Utilize an angle plate and surface plate for accuracy.

### **5. Punching Dots:**

- Dots are punched along the scribed lines using a dot punch, ensuring precision in the markings.

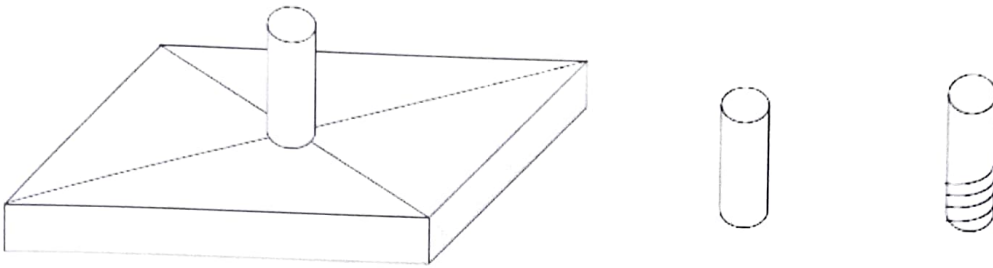
### **6. Filing the Remaining Sides:**

- The workpiece is refitted in the bench vice, and the remaining two sides are filed.
- Check the flatness of the surfaces after filing. As filing removes material more precisely and gradually than sawing, it is preferred for this final shaping.

### **Result:**

- A square piece of mild steel with specified dimension is successfully obtained by following the above procedure.

**Aim 3:** To centre punch the filed Mild Steel piece, and make internal and external threading as shown in Figure.



#### **Tools Required:**

- **Bench Vice:** For securely holding the workpiece during filing, marking, drilling, and threading.
- **Set of Files:** For squaring and smoothing the surfaces of the mild steel piece.
- **Steel Rule:** For accurate measurement and marking.
- **Try-Square:** For verifying the squareness of the workpiece and ensuring right angles.
- **Vernier Caliper:** For precise measurement of dimensions and thread diameters.
- **Vernier Height Gauge:** For finding and marking the exact center of the workpiece.
- **Ball-Peen Hammer:** For striking the dot punch and center punch.
- **Scriber:** For scribing lines on the workpiece to indicate cutting or drilling points.
- **Dot Punch:** For creating an initial small indentation at the center mark.
- **Centre Punch:** For enlarging the indentation to guide the drill bit.

#### **Procedure**

##### **1. Square Filing the Workpiece:**

- Secure the mild steel piece in a bench vice.
- Use a set of files to ensure all four faces of the steel piece are square and smooth.
- Measure and verify the squareness using a try-square. Adjust filing if necessary to achieve precise right angles.

## **2. Marking the Centre:**

- After filing, place the square workpiece on a flat surface.
- Use a steel rule and vernier height gauge to find the center of the workpiece.
- Scribe a line along the center using a scribe to mark the position.
- Cross-check the center mark with a try-square for accuracy.

## **3. Punching the Centre Mark:**

- Place the workpiece back in the bench vice, ensuring it is stable.
- Position the dot punch precisely on the scribed center mark.
- Lightly tap the dot punch with a ball-peen hammer to create a small indentation.
- Follow up with a centre punch placed over the dot punch mark, and strike it firmly with the hammer to deepen the indentation. This ensures accurate drilling.

## **4. Drilling the Hole:**

- Secure the workpiece on a drilling machine table using clamps.
- Align the drill bit with the centre-punched mark.
- Start the drilling machine and carefully drill a hole of the required diameter.
- Ensure that the hole depth is appropriate for the length of the cylindrical rod that will be threaded and inserted.

## **5. Internal Threading:**

- After drilling, the hole is prepared for internal threading.
- Select the correct tap size corresponding to the hole diameter.
- Attach the tap to a tap wrench and apply cutting fluid to reduce friction.
- Insert the tap into the hole and rotate it clockwise to cut the threads inside the hole.
- After completing the threading, reverse the tap to remove it, clearing any debris.

## **6. External Threading:**

- Secure the cylindrical rod in the bench vice.
- Select the appropriate die corresponding to the external diameter of the rod.

- Place the die in a die holder and apply cutting fluid to the rod.
- Position the die at the end of the rod and rotate the die holder clockwise to start cutting the external threads.
- Continue threading until the desired length is fully threaded.
- Reverse the die holder to remove it, ensuring a clean, well-formed thread.

#### 7. **Fitting the Threaded Rod:**

- Test the fit by screwing the externally threaded rod into the internally threaded hole.
- Ensure that the threads are smooth, and the rod fits securely without excessive play.

#### 8. **Final Inspection:**

- Check the alignment and fit of the threaded rod in the hole.
- Ensure that the threads are clean and free of any burrs or imperfections.

#### 9. **Select the Appropriate Thread Pitch Gauge:**

- Choose a thread pitch gauge that corresponds to the type of thread you're measuring (e.g., metric or imperial).
- Match the gauge's teeth to the thread. The correct gauge will fit snugly into the thread grooves without gaps or rocking.

#### 10. **Measure the Thread Pitch:**

- Once you've selected the right gauge, read the pitch value marked on the gauge (e.g., 1.5 mm, 20 TPI).
- For metric threads, the pitch is the distance between two adjacent threads in millimeters.
- For imperial threads, the pitch is measured in threads per inch (TPI).



## 1. FITTING

### 1.1. INTRODUCTION

Machine tools are capable of producing work at a faster rate, but, there are occasions when components are processed at the bench. Sometimes, it becomes necessary to replace or repair component which must be fit accurately with another component on reassembly. This involves a certain amount of hand fitting. The assembly of machine tools, jigs, gauges, etc, involves certain amount of bench work. The accuracy of work done depends upon the experience and skill of the fitter.

The term 'bench work' refers to the production of components by hand on the bench, where as fitting deals with the assembly of mating parts, through removal of metal, to obtain the required fit.

Both the bench work and fitting requires the use of number of simple hand tools and considerable manual efforts. The operations in the above works consist of filing, chipping, scraping, sawing drilling, and tapping.

### 1.2. HOLDING TOOLS

#### Bench vice

The bench vice is a work holding device. It is the most commonly used vice in a fitting shop. The bench vice is shown in Figure 1.1.

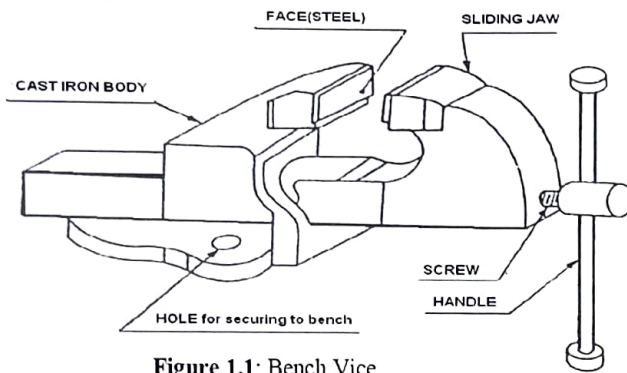


Figure 1.1: Bench Vice

It is fixed to the bench with bolts and nuts. The vice body consists of two main parts, fixed jaw and movable jaw. When the vice handle is turned in a clockwise direction, the sliding jaw forces the work against the fixed jaw. Jaw plates are made of hardened steel.

Serrations on the jaws ensure a good grip. Jaw caps made of soft material are used to protect finished surfaces, gripped in the vice. The size of the vice is specified by the length of the jaws.

The vice body is made of cast Iron which is strong in compression, weak in tension and so fractures under shocks and therefore should never be hammered.

#### V-block

V-block is rectangular or square block with a V groove on one or both sides opposite to each other. The angle of the 'V' is usually  $90^\circ$ . V-block with a clamp is used to hold cylindrical work securely, during layout of measurement, for measuring operations or for drilling for this the bar is faced longitudinally in the V groove and the screw of V-clamp is tightened. This grip the rod is firm with its axis parallel to the axis of the V groove.



## C-Clamp

This is used to hold work against an angle plate or v-block or any other surface, when gripping is required.

Its fixed jaw is shaped like English alphabet 'C' and the movable jaw is round in shape and directly fitted to the threaded screw at the end. The working principle of this clamp is the same as that of the bench vice.

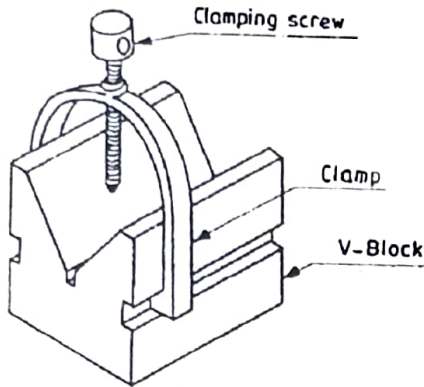


Figure 1.2: V-block

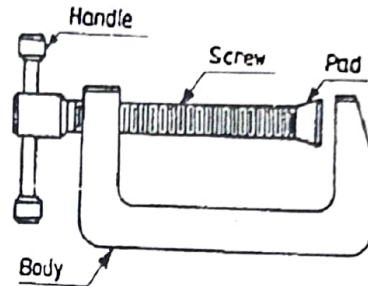


Figure 1.3: C-clamp

## 1.3 MARKING AND MEASURING TOOLS

### Surface plate

The surface plate is machined to fine limits and is used for testing the flatness of the work piece. It is also used for marking out small box and is more precious than the marking table. The degree of the finished depends upon whether it is designed for bench work in a fitting shop or for using in an inspection room; the surface plate is made of Cast Iron, hardened Steel or Granite stone. It is specified by length, width, height and grade. Handles are provided on two opposite sides, to carry it while shifting from one place to another.

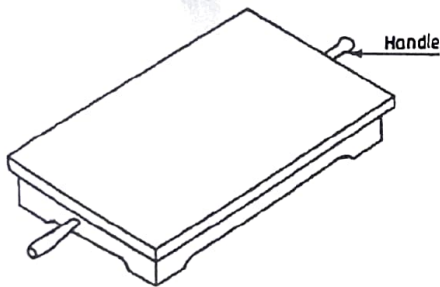


Figure 1.4: Surface plate

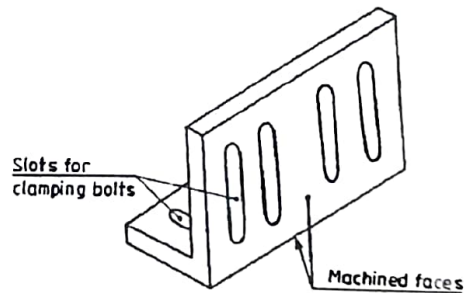


Figure 1.5: Angle plate

### Try square

It is measuring and marking tool for  $90^\circ$  angle. In practice, it is used for checking the squareness of many types of small works when extreme accuracy is not required. The blade of the Try square is made of hardened steel and the stock of cast Iron or steel. The size of the Try square is specified by the length of the blade.

### Scriber

A Scriber is a slender steel tool, used to scribe or mark lines on metal work pieces. It is made of

hardened and tempered High Carbon Steel. The Tip of the scribe is generally ground at  $12^{\circ}$  to  $15^{\circ}$ . It is generally available in lengths, ranging from 125mm to 250mm. It has two pointed ends the bent end is used for marking lines where the straight end cannot reach.

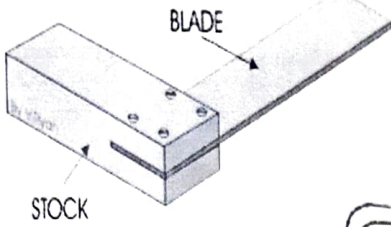


Figure 1.6: Try square

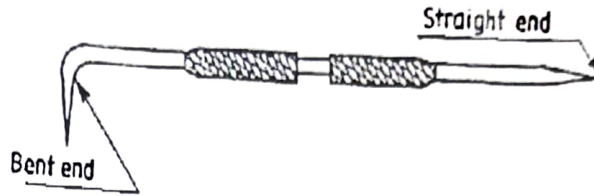


Figure 1.7: Scriber

### Odd leg Caliper

This is also called 'Jenny Caliper' or Hermaphrodite. This is used for marking parallel liners from a finished edge and also for locating the center of round bars; it has one leg pointed like a divider and the other leg bent like a caliper. It is specified by the length of the leg up to the hinge point.

### Divider

It is basically similar to the calipers except that its legs are kept straight and pointed at the measuring edge. This is used for marking circles, arcs laying out perpendicular lines, by setting lines. It is made of case hardened mild steel or hardened and tempered low carbon steel. Its size is specified by the length of the leg.

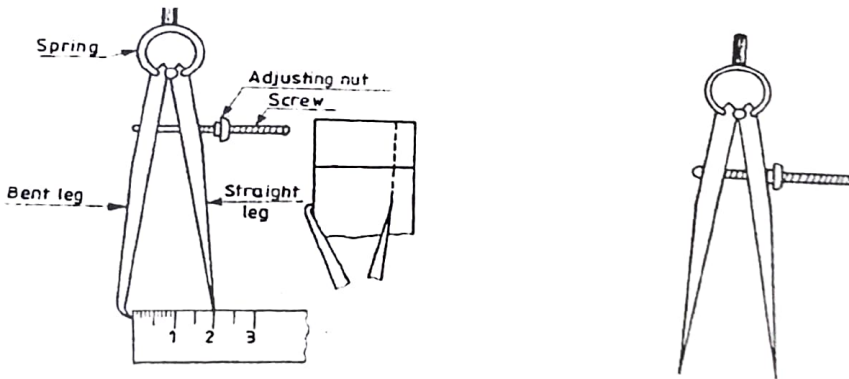


Figure 1.8: Odd leg caliper and divider

## Trammel

Trammel is used for drawing large circles or arcs.

## Punches

These are used for making indentations on the scribed lines, to make them visible clearly. These are made of high carbon steel.

A punch is specified by its length and diameter (say as 150' 12.5mm). It consists of a cylindrical knurled body, which is plain for some length at the top of it. At the other end, it is ground to a point. The tapered point of the punch is hardened over a length of 20 to 30mm.

Dot punch is used to lightly indent along the layout lines, to locate center of holes and to provide a small center mark for divider point, etc. for this purpose, the punch is ground to a conical point having  $60^\circ$  included angle.

Centre punch is similar to the dot punch, except that it is ground to a conical point having  $90^\circ$  included angle. It is used to mark the location of the holes to be drilled.

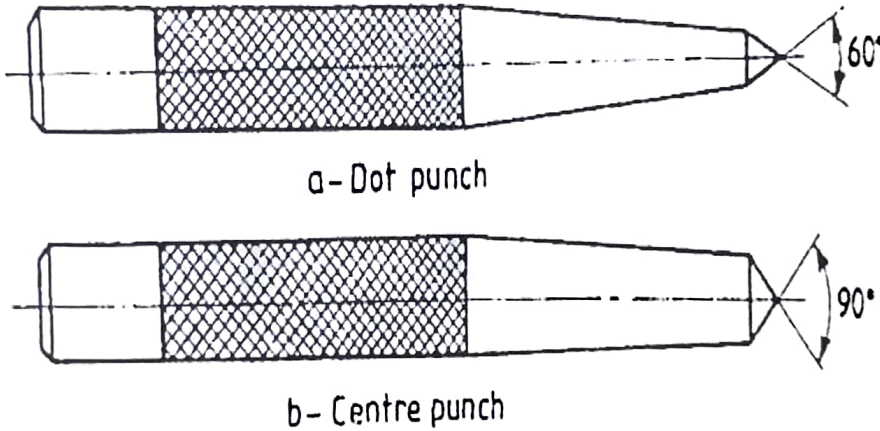


Figure 1.9: Punches

Calipers

They are indirect measuring tools used to measure or transfer linear dimensions. These are used with the help of a steel Rule to check inside and outside measurements. These are made of Case hardened mild steel or hardened and tempered low carbon steel. While using, but the legs of the caliper are set against the surface of the work, whether inside or outside and the distance between the legs is measured with the help of a scale and the same can be transferred to another desired place. These are specified by the length of the leg. In the case of outside caliper, the legs are bent inwards and in the case of inside caliper, the legs bent outwards.

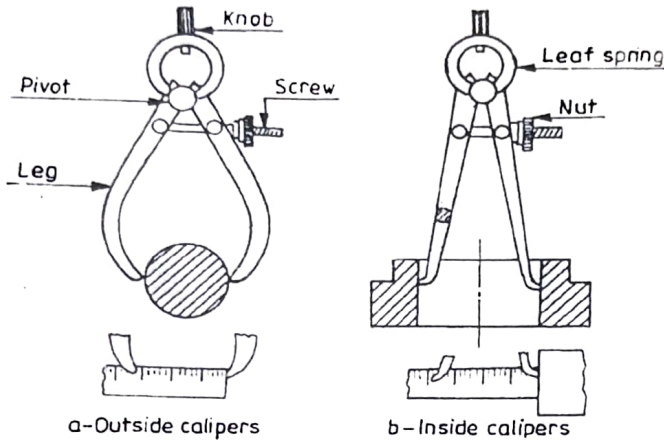


Figure 1.10: Calipers

Vernier Calipers

These are used for measuring outside as well as inside dimensions accurately. It may also be used as a depth gauge. It has two jaws. One jaw is formed at one end of its main scale and the other jaw is made part of a vernier scale.

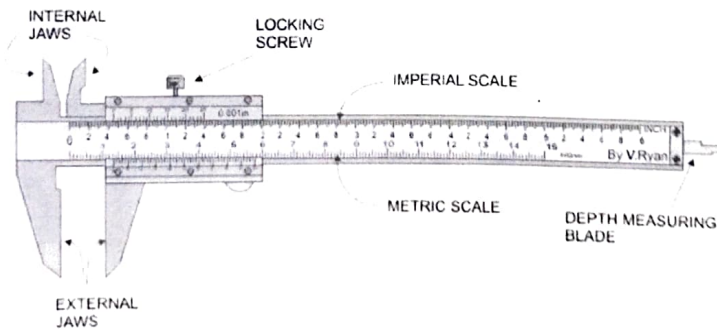


Figure 1.11: Vernier caliper



## Vernier Height Gauge

The Vernier Height gauge clamped with a scriber. It is used for Lay out work and offset scriber is used when it is required to take measurement from the surface, on which the gauge is standing. The accuracy and working principle of this gauge are the same as those of the vernier calipers. It's size is specified by the maximum height that can be measured by it. It is made of Nickel Chromium Steel.

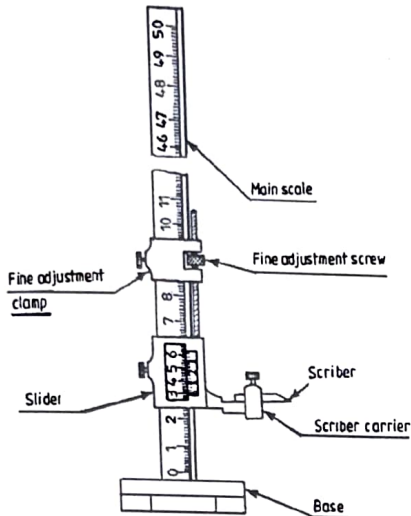


Figure 1.12: Vernier Height gauge

## 1.4 CUTTING TOOLS

### Hack Saw

The Hack Saw is used for cutting metal by hand. It consists of a frame, which holds a thin blade, firmly in position. Hacksaw blade is specified by the number of teeth for centimeter. Hacksaw blades have a number of teeth ranging from 5 to 15 per centimeter (cm). Blades having lesser number of teeth per cm are used for cutting soft materials like aluminum, brass and bronze. Blades having larger number of teeth per centimeter are used for cutting hard materials like steel and cast Iron.

Hacksaw blades are classified as (i) All hard and (ii) flexible type. The all hard blades are made of H.S.S, hardened and tempered throughout to retain their cutting edges longer. These are used to cut hard metals. These blades are hard and brittle and can break easily by twisting and forcing them into the work while sawing. Flexible blades are made of H.S.S or low alloy steel but only the teeth are hardened and the rest of the blade is soft and flexible. These are suitable for use by unskilled or semiskilled persons.

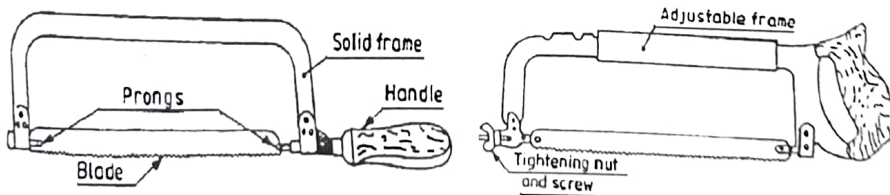


Figure 1.13: Hacksaw frame with blade

The teeth of the hacksaw blade are staggered, as shown in figure and known as a 'set of teeth'. These make slots wider than the blade thickness, preventing the blade from jamming.



Figure 1.14: Set of teeth

### Chisels

Chisels are used for removing surplus metal or for cutting thin sheets. These tools are made from 0.9% to 1.0% carbon steel of octagonal or hexagonal section. Chisels are annealed, hardened and tempered to produce a tough shank and hard cutting edge. Annealing relieves the internal stresses in a metal. The cutting angle of the chisel for general purpose is about  $60^\circ$ .

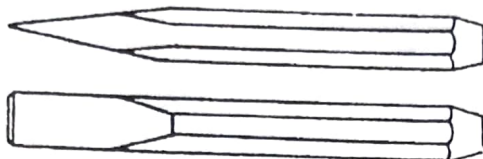


Figure 1.15: Flat chisel

### Twist Drill

Twist drills are used for making holes. These are made of High speed steel. Both straight and taper shank twist drills are used. The parallel shank twist drill can be held in an ordinary self-centering drill chuck. The taper shank twist drill fits into a corresponding tapered bore provided in the drilling machine spindle.

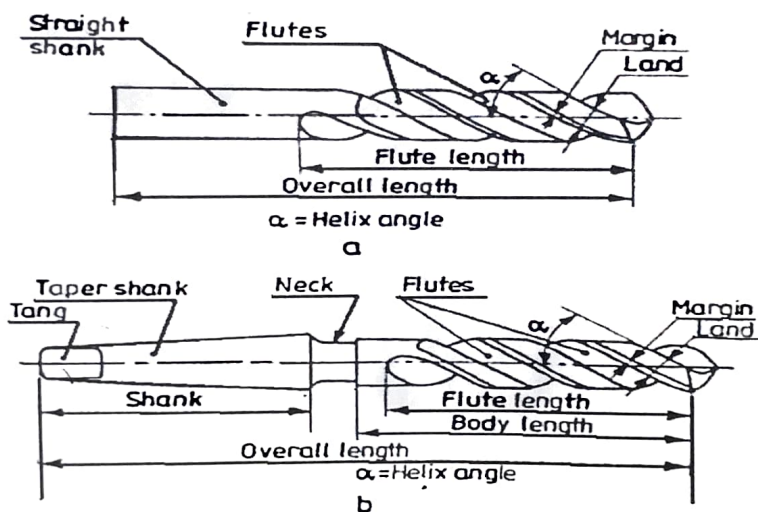


Figure 1.16: Twist drills

### Taps and Tap wrenches

A tap is a hardened and steel tool, used for cutting internal thread in a drill hole. Hand Taps are usually supplied in sets of three in each diameter and thread size. Each set consists of a taper tap, intermediate tap and plug or bottoming tap. Taps are made of high carbon steel or high speed steel.



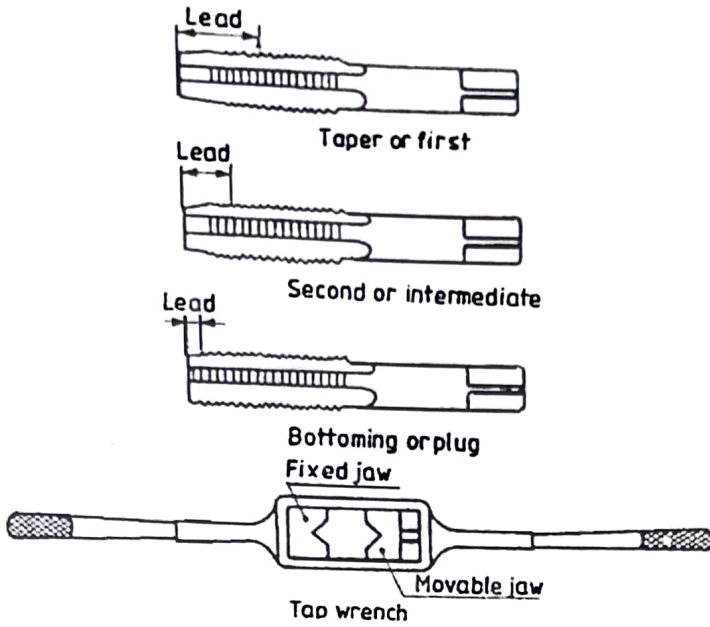


Figure Taps and tap wrench

#### Dies and die holders

Dies are the cutting tools used for making external thread. Dies are made either solid or split type. They are fixed in a die stock for holding and adjusting the die gap. They are made of Steel or High Carbon Steel.

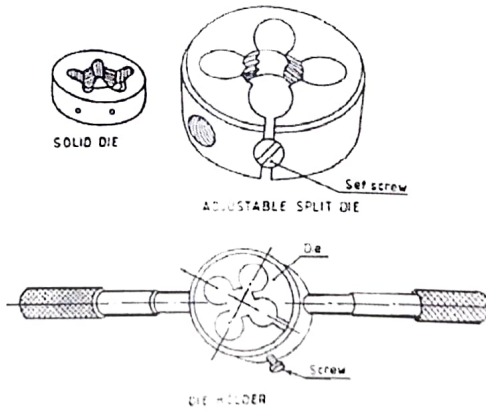


Figure 1.18: Dies and die holder

## Bench Drilling Machine

Holes are drilled for fastening parts with rivets, bolts or for producing internal thread. Bench drilling machine is the most versatile machine used in a fitting shop for the purpose. Twist drills, made of tool steel or high speed steel are used with the drilling machine for drilling holes.

Following are the stages in drilling work

1. Select the correct size drills, put it into the chuck and lock it firmly
2. Adjust the speed of the machine to suit the work by changing the belt on the pulleys. Use high speed for small drills and soft materials and low speed for large diameter drills and hard materials.
3. Layout of the location of the hole and mark it with a centre punch.
4. Hold the work firmly in the vice on the machine table and clamp it directly on to the machine table.
6. Put on the power, locate the punch mark and apply slight pressure with the Feed Handle. Once drilling is commenced at the correct location, apply enough pressure and continue drilling. When drilling steel apply cutting oil at the drilling point.
7. Release the pressure slightly, when the drill point pierces the lower surface of the metal. This prevents the drill catching and damaging the work or drill.
8. On completion of drilling retract the drill out of the work and put off the power supply.

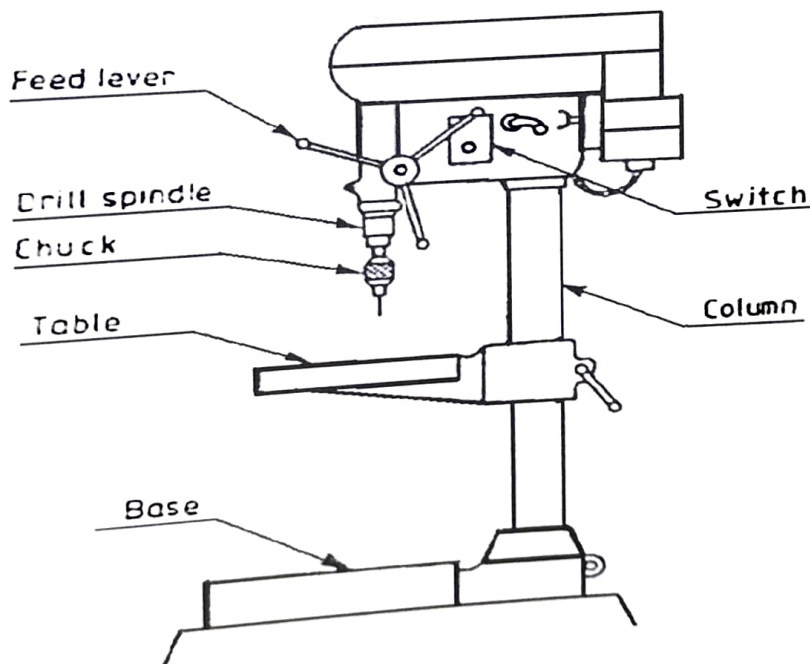


Figure 1.19: Bench drill

## 1.5 FINISHING TOOLS

### Reamers

Reaming is an operation of sizing and finishing a drilled hole, with the help of a cutting tool called reamer having a number of cutting edges. For this, a hole is first drilled, the size of which is slightly smaller than the finished size and then a hand reamer or machine reamer is used for finishing the hole to the correct size.

Hand Reamer is made of High Carbon Steel and has left hand spiral flutes so that, it is prevented from screwing into the whole during operation. The Shank end of the reamer is made straight of the reamer and with the work piece held in the vice. The body of the reamer is given a slight taper at its working end, for its easy entry into the whole during operation, it is rotated only in clock wise direction and also while removing it from the whole.

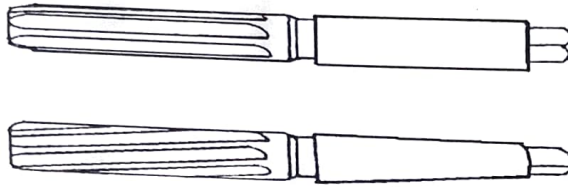


Figure 1.20: Reamers

### Files

Filing is one of the methods of removing small amounts of material from the surface of a metal part. A file is hardened steel too, having small parallel rows of cutting edges or teeth on its surfaces.

On the faces, the teeth are usually diagonal to the edge. One end of the file is shaped to fit into a wooden handle. The figure shows various parts of a hand file. The hand file is parallel in width and tapering slightly in thickness, towards the tip. It is provided with double cut teeth. On the faces, single cut on one edge and no teeth on the other edge, which is known as a safe edge.

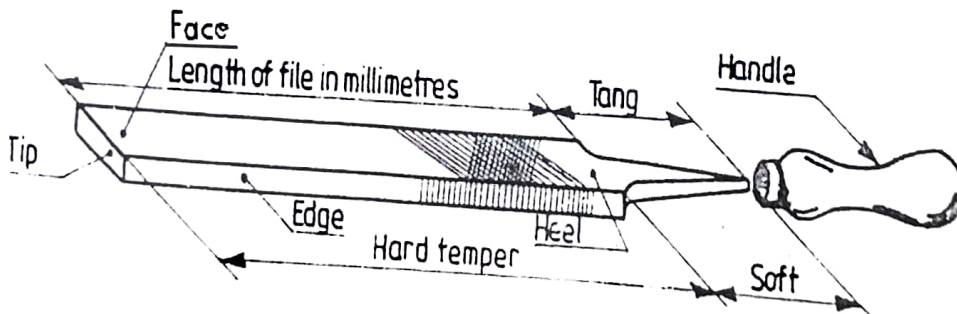


Figure 1.21: Parts of a hand file

Files are classified according to their shape, cutting teeth and pitch or grade of the teeth. The figure shows the various types of files based on their shape.

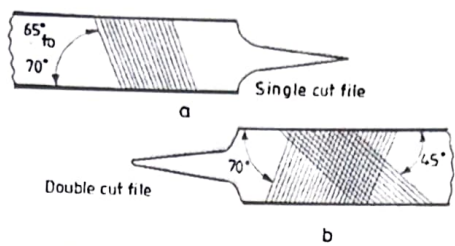


Figure 1.22: Single and double cut files

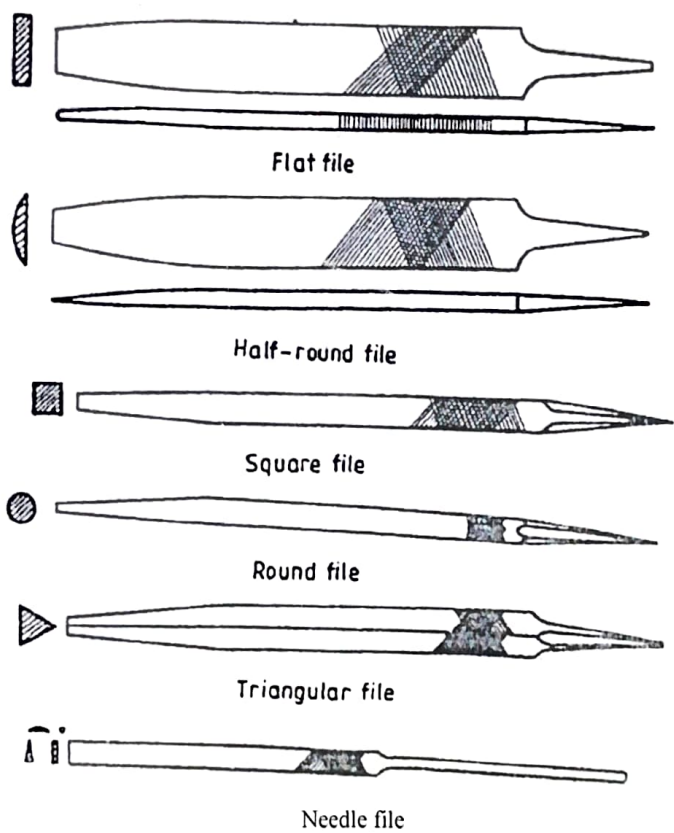


Figure 1.23: Types of files

## 1.6 MISCELLANEOUS TOOLS

### File card

It is a metal brush, used for cleaning the files, to free them from filings, clogged in between the teeth.

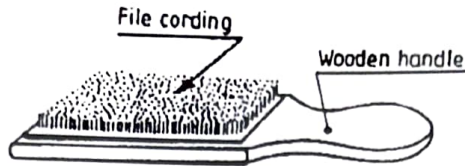


Figure 1.24: File card

### Spirit level

It is used to check the leveling of machines.

### Ball Peen Hammer

Ball Peen Hammers are named, depending upon their shape and material and specified by their weight. A ball peen hammer has a flat face which is used for general work and a ball end, particularly used for riveting.

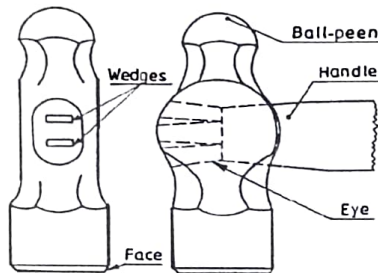


Figure 1.25: Ball peen hammer

### 1. Cross Peen Hammer

It is similar to ball peen hammer, except the shape of the peen. This is used for chipping, riveting, bending and stretching metals and hammering inside the curves and shoulders.

### Straight Peen Hammer

This is similar to cross peen hammer, but its peen is in line with the hammer handle. It is used for swaging, riveting in restricted places and stretching metals.

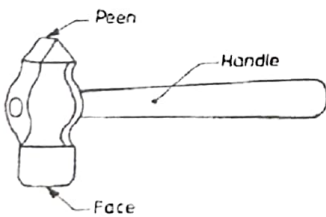


Figure 1.26: Cross peen hammer

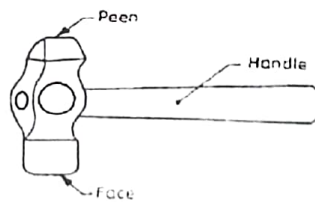


Figure 1.27: Straight peen hammer



### Screw driver

A screw driver is designed to turn screws. The blade is made of steel and is available in different lengths and diameters. The grinding of the tip to the correct shape is very important.

A star screw driver is specially designed to fit the head of star screws. The end of the blade is fluted instead of flattened. The screw driver is specified by the length of the metal part from handle to the tip.

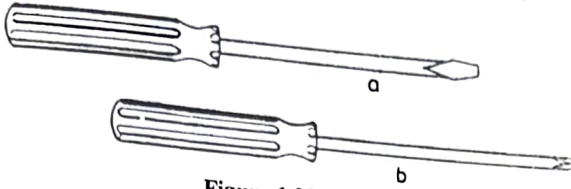


Figure 1.28: Screw drivers

### Spanners

A spanner or wrench is a tool for turning nuts and bolts. It is usually made of forged steel. There are many kinds of spanners. They are named according to the application. The size of the spanner denotes the size of the bolt on which it can work.

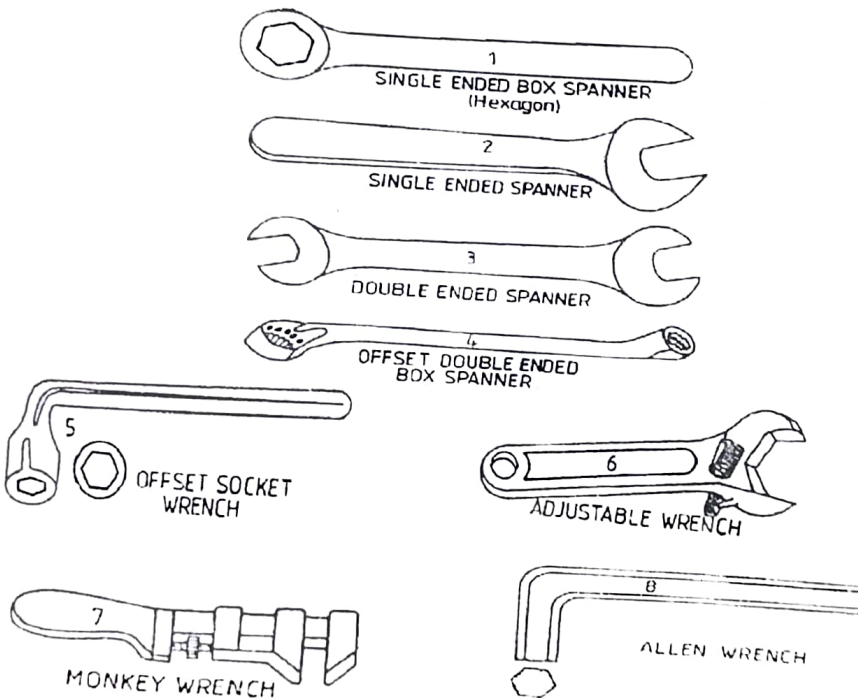


Figure 1.28: Spanners